

EXPRIMENT - 2

AIM: Convert the ER diagram into a relational schema. Write complete MySQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

SOURCE CODE:

```
CREATE DATABASE ECOMMERCE;  
USE ECOMMERCE;
```

```
CREATE TABLE CUSTOMER(  
CustomerID INT PRIMARY KEY,  
Name VARCHAR(50) NOT NULL,  
Email VARCHAR(100) NOT NULL UNIQUE,  
Password VARCHAR(100) NOT NULL,  
DateOfBirth DATE NOT NULL  
);
```

```
CREATE TABLE CUSTOMER_PHONE(  
CustomerID INT,  
Phone VARCHAR(15),  
PRIMARY KEY(CustomerID,Phone),  
FOREIGN KEY(CustomerID) REFERENCES CUSTOMER(CustomerID)  
ON DELETE CASCADE  
);
```

```
CREATE TABLE CUSTOMER_ALT_PHONE(  
CustomerID INT,  
AltPhone VARCHAR(15),  
PRIMARY KEY(CustomerID,AltPhone),  
FOREIGN KEY(CustomerID) REFERENCES CUSTOMER(CustomerID)  
ON DELETE CASCADE  
);
```

```
CREATE TABLE SELLER(  
SellerID INT PRIMARY KEY,  
SellerName VARCHAR(50) NOT NULL,  
Email VARCHAR(100) UNIQUE NOT NULL,  
GSTIN VARCHAR(20) UNIQUE NOT NULL,  
AccountNo VARCHAR(20),  
IFSCCode VARCHAR(15),  
BankName VARCHAR(50)  
);
```

```
CREATE TABLE SELLER_PHONE(  
SellerID INT,  
Phone VARCHAR(15),  
PRIMARY KEY(SellerID,Phone),  
FOREIGN KEY(SellerID) REFERENCES SELLER(SellerID)  
ON DELETE CASCADE  
);
```

```
CREATE TABLE CATEGORY(  
CategoryID INT PRIMARY KEY,  
CategoryName VARCHAR(50) NOT NULL,  
Description VARCHAR(100),  
ParentCategoryID INT,  
FOREIGN KEY(ParentCategoryID) REFERENCES CATEGORY(CategoryID)  
ON DELETE SET NULL  
);
```

```
CREATE TABLE PRODUCT(  
ProductID INT PRIMARY KEY,  
ProductName VARCHAR(50) NOT NULL,  
Description VARCHAR(100),  
Price DECIMAL(10,2) NOT NULL,  
StockQty INT NOT NULL,  
Weight DECIMAL(8,2),  
SellerID INT NOT NULL,  
CategoryID INT,  
FOREIGN KEY(SellerID) REFERENCES SELLER(SellerID)  
ON DELETE CASCADE,  
FOREIGN KEY(CategoryID) REFERENCES CATEGORY(CategoryID)  
ON DELETE SET NULL  
);
```

```
CREATE TABLE PRODUCT_IMAGE(  
ProductID INT,  
Image VARCHAR(200),  
PRIMARY KEY(ProductID,Image),  
FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)  
ON DELETE CASCADE  
);
```

```
CREATE TABLE ELECTRONICS(  
ProductID INT PRIMARY KEY,  
Brand VARCHAR(50),  
Warranty VARCHAR(30),  
PowerRating VARCHAR(30),  
FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)  
ON DELETE CASCADE  
);
```

```
CREATE TABLE CLOTHING(  
ProductID INT PRIMARY KEY,  
Size VARCHAR(20),  
Colour VARCHAR(20),  
Material VARCHAR(30),  
FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)  
ON DELETE CASCADE  
);
```

```
CREATE TABLE BOOKS(  
ProductID INT PRIMARY KEY,  
ISBN VARCHAR(20) UNIQUE,  
Author VARCHAR(50),  
Publisher VARCHAR(50),
```

```
FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
ON DELETE CASCADE
);
```

```
CREATE TABLE ADDRESS(
AddressID INT PRIMARY KEY,
DoorNo VARCHAR(10),
Street VARCHAR(50),
Area VARCHAR(50),
City VARCHAR(30),
State VARCHAR(30),
Pincode VARCHAR(10) NOT NULL,
AddressType VARCHAR(20),
CustomerID INT NOT NULL,
FOREIGN KEY(CustomerID) REFERENCES CUSTOMER(CustomerID)
ON DELETE CASCADE
);
```

```
CREATE TABLE ADDRESS_MOBILE(
AddressID INT,
MobileNo VARCHAR(15),
PRIMARY KEY(AddressID,MobileNo),
FOREIGN KEY(AddressID) REFERENCES ADDRESS(AddressID)
ON DELETE CASCADE
);
```

```
CREATE TABLE ORDERS(
OrderID INT PRIMARY KEY,
OrderDate DATE NOT NULL,
OrderStatus VARCHAR(20) NOT NULL,
TotalAmount DECIMAL(10,2) NOT NULL,
CustomerID INT NOT NULL,
AddressID INT,
FOREIGN KEY(CustomerID) REFERENCES CUSTOMER(CustomerID)
ON DELETE CASCADE,
FOREIGN KEY(AddressID) REFERENCES ADDRESS(AddressID)
ON DELETE SET NULL
);
```

```
CREATE TABLE ORDER_COUPON(
OrderID INT,
CouponCode VARCHAR(30),
PRIMARY KEY(OrderID,CouponCode),
FOREIGN KEY(OrderID) REFERENCES ORDERS(OrderID)
ON DELETE CASCADE
);
```

```
CREATE TABLE ORDERITEM(
OrderID INT,
ItemNo INT,
ProductID INT NOT NULL,
ItemPrice DECIMAL(10,2) NOT NULL,
Quantity INT NOT NULL,
Discount DECIMAL(10,2),
TaxAmount DECIMAL(10,2),
```

```
PRIMARY KEY(OrderID,ItemNo),
FOREIGN KEY(OrderID) REFERENCES ORDERS(OrderID)
ON DELETE CASCADE,
FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
ON DELETE CASCADE
);
```

```
CREATE TABLE PAYMENT(
PaymentID INT PRIMARY KEY,
PaymentDate DATE NOT NULL,
Amount DECIMAL(10,2) NOT NULL,
PaymentMethod VARCHAR(30) NOT NULL,
TransactionID VARCHAR(30) UNIQUE,
PaymentStatus VARCHAR(20) NOT NULL,
OrderID INT UNIQUE NOT NULL,
FOREIGN KEY(OrderID) REFERENCES ORDERS(OrderID)
ON DELETE CASCADE
);
```

```
CREATE TABLE DELIVERY(
DeliveryID INT PRIMARY KEY,
DeliveryDate DATE,
DeliveryStatus VARCHAR(20) NOT NULL,
TrackingID VARCHAR(30) UNIQUE,
DeliveryPartner VARCHAR(50),
ShippingCharge DECIMAL(8,2),
OrderID INT UNIQUE NOT NULL,
AddressID INT,
FOREIGN KEY(OrderID) REFERENCES ORDERS(OrderID)
ON DELETE CASCADE,
FOREIGN KEY(AddressID) REFERENCES ADDRESS(AddressID)
ON DELETE SET NULL
);
```

```
INSERT INTO CUSTOMER VALUES
(1,'Karan Kushwaha','karankushwaha18210@gmail.com','pass123','2006-03-08');
```

```
INSERT INTO CUSTOMER_PHONE VALUES
(1,'8953318721');
```

```
INSERT INTO CUSTOMER_ALT_PHONE VALUES
(1,'7307598186');
```

```
INSERT INTO SELLER VALUES
(101,'TechStore','tech@gmail.com','GST123','123456','SBIN001','SBI');
```

```
INSERT INTO SELLER_PHONE VALUES
(101,'9621387818');
```

```
INSERT INTO CATEGORY VALUES
(201,'Electronics','Electronic items',NULL);
```

```
INSERT INTO CATEGORY VALUES
(202,'Mobiles','Mobile phones',201);
```

INSERT INTO PRODUCT VALUES
(301,'Laptop','Gaming Laptop',75000,10,2.5,101,201);

INSERT INTO ELECTRONICS VALUES
(301,'ASUS','2 Years','180W');

INSERT INTO PRODUCT_IMAGE VALUES
(301,'laptop.jpg');

INSERT INTO ADDRESS VALUES
(401,'12','Eldeco','Transport Nagar','Lucknow','Uttar Pradesh','160017','Home',1);

INSERT INTO ADDRESS_MOBILE VALUES
(401,'9876543210');

INSERT INTO ORDERS VALUES
(501,'2026-08-14','Placed',75000,1,401);

INSERT INTO ORDERITEM VALUES
(501,1,301,75000,1,1000,9000);

INSERT INTO ORDER_COUPON VALUES
(501,'WELCOME10');

INSERT INTO PAYMENT VALUES
(601,'2026-08-14',74000,'UPI','TXN123','Paid',501);

INSERT INTO DELIVERY VALUES
(701,NULL,'Processing','TRK123','Delhivery',100,501,401);

SELECT * FROM CUSTOMER;
SELECT * FROM CUSTOMER_PHONE;
SELECT * FROM CUSTOMER_ALT_PHONE;
SELECT * FROM SELLER;
SELECT * FROM SELLER_PHONE;
SELECT * FROM CATEGORY;
SELECT * FROM PRODUCT;
SELECT * FROM PRODUCT_IMAGE;
SELECT * FROM ELECTRONICS;
SELECT * FROM CLOTHING;
SELECT * FROM BOOKS;
SELECT * FROM ADDRESS;
SELECT * FROM ADDRESS_MOBILE;
SELECT * FROM ORDERS;
SELECT * FROM ORDER_COUPON;
SELECT * FROM ORDERITEM;
SELECT * FROM PAYMENT;
SELECT * FROM DELIVERY;

OUTPUT:

 KARAN KUSHWAHA

I/O AI Agent 476 ms 

STDIN
Input for the program (Optional)

Output:

CustomerID	Name	Email	Password	DateOfBirth
1	Karan Kushwaha	karankushwaha18210@gmail.com	pass123	2006-03-08

CustomerID	Phone
1	8953318721

CustomerID	AltPhone
1	7307598186

SellerID	SellerName	Email	GSTIN	AccountNo	IFSCCode	BankName
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Light MySQL

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STDIN
Input for the program (Optional)

Output:

SellerID	SellerName	Email	GSTIN	AccountNo	IFSCCode	BankName
101	TechStore	tech@gmail.com	GST123	123456	SBIN001	SBI

SellerID	Phone
101	9621387818

CategoryID	CategoryName	Description	ParentCategoryID
201	Electronics	Electronic items	NULL
202	Mobiles	Mobile phones	201

ProductID	ProductName	Description	Price	StockQty	Weight	SellerID	CategoryID
301	Laptop	Gaming Laptop	75000.00	10	2.50	101	201

Light MySQL

Output:

OrderID	OrderDate	OrderStatus	TotalAmount	CustomerID	AddressID
501	2026-08-14	Placed	75000.00	1	401

OrderID	CouponCode
501	WELCOME10

OrderID	ItemNo	ProductID	ItemPrice	Quantity	Discount	TaxAmount
501	1	301	75000.00	1	1000.00	9000.00

PaymentID	PaymentDate	Amount	PaymentMethod	TransactionID	PaymentStatus	OrderID
601	2026-08-14	74000.00	UPI	TXN123	Paid	501

DeliveryID	DeliveryDate	DeliveryStatus	TrackingID	DeliveryPartner	ShippingCharge	OrderID	AddressID
701	NULL	Processing	TRK123	Delhivery	100.00	501	401